

# ANDRY PAEZ

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## EXPERIENCE

### Research Assistant, Dept. of Physics and Astronomy

Feb 2026 – Present

San Jose State University (advisor: Dr. Hilary Hurst), 30 hrs/week

- Develop spinor Bose-Einstein condensate simulations (Gross-Pitaevskii equation); 150+ commits merged into the research group's codebase through advisor-reviewed pull requests.
- Vectorized the core kinetic-energy operator for up to 1,880x speedup (161 s to 0.09 s) with bit-exact numerical parity verified by a benchmark harness.
- Built the data pipeline: CLI with HDF5 output, hash-addressed run catalog with rebuildable SQLite index, and a Slurm write-then-index HPC workflow that avoids concurrent-writer contention.
- Designed controlled storage benchmarks and authored the store-vs-recompute recommendation adopted as the project default.

## PROJECTS

### Reachy Mini Robot Control Console

Jun 2026 – Present

- Built a FastAPI web control console for a Reachy Mini robot on a headless Jetson Orin Nano, designed as a pure HTTP/WebSocket proxy to the robot daemon; 162 tests passing, functionally verified against live hardware.
- Built a voice assistant pipeline on the same platform: microphone to faster-whisper to LLM (with live camera frames for visual questions) to TTS; 286 tests passing.
- Root-caused failures across the stack: compiled GStreamer webrtc sink from source for arm64 with NVMM buffer transport, traced speaker silence to ALSA card remapping after an OS migration, and fixed a SIGINT-forwarding bug with a process-tree signaller.
- Reduced idle UI overhead to zero DOM writes per tick (change-guarded rendering, SSE rate reduction), fixing a measured phone battery/heat problem; verified by driving the served app with Playwright.

### LocalLM, Agentic Desktop App for Local Models

Apr 2026

- Built a cross-platform desktop app for local Ollama models with an agentic tool-calling layer (web search, file I/O, code execution) in Rust (Tauri 2) with a React/TypeScript frontend and streaming chat.
- Implemented a 47-model recommendation registry with RAM/disk metadata and machine-spec detection for quantization-aware model selection.

### omnisense, Offline Voice and Vision Assistant for Jetson

Apr 2026

- Built a modular multimodal assistant package: camera abstraction (GStreamer/OpenCV/mock backends), faster-whisper speech-to-text, voice activity detection, YOLO overlay, FastAPI REST/WebSocket interface; testable without hardware via mock backends.

### Sage, Local-First Knowledge Agent

Feb 2025 – May 2026

- Designed and built a desktop knowledge agent over 15 months (130 commits): document ingestion into a local SQLite store with embeddings, grounded chat with citations, quiz/report/audio generation; Tauri v2, Next.js, FastAPI, multi-provider LLM abstraction (OpenAI, Anthropic, Ollama), CI.

## OPEN SOURCE

- **NousResearch/hermes-agent**: two merged bugfixes to a large open-source agent framework (systemd gateway restart race; Discord 8KB command-payload split).

## SKILLS

- **AI Systems**: local LLM inference (llama.cpp, Ollama), agent architectures and tool-calling loops, RAG, MCP, quantization-aware model selection, edge deployment (Jetson Orin Nano), faster-whisper
- **Languages**: Python (primary), TypeScript/JavaScript, Rust, C/C++, SQL, Bash
- **Systems**: FastAPI, WebSockets/SSE, Tauri, pytest, GitHub Actions CI, Docker, Playwright, Linux (systemd, GStreamer, ALSA/PipeWire), Tailscale, Git; multi-agent coding workflows with verification-first review gates

## EDUCATION

### B.S., Physics

May 2026

San Jose State University. GPA 3.80. Dean's Scholar (Fall 2024, Fall 2025)